

FRONTEND AI UX PATTERNS

Module 5 — Governed AI Feature Delivery

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Now: build structured, reviewable AI UX patterns.

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The frontend is a control layer. Design it like one.

UI PATTERNS THAT WORK

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- Confidence and evidence cues shown in context
- Editable review step before commit
- Streaming states with clear progression

CHAT-FIRST VS TASK-FIRST UX

Approach	Strength	Risk
Chat-first UI	Fast prototyping	Weak structure, poor verification
Task-first structured UI	Clear review and control	More upfront design effort

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- Separate extracted data from explanatory or supporting notes.
- Show status per field: accepted, uncertain, or missing.
- Keep the frontend response shape aligned to the backend contract.

If the UI contract diverges from the backend contract, both become harder to govern.

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Hiding uncertainty does not remove it. It just removes the reviewer's ability to act on it.

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- Do not expose internal prompt text or policy internals.

REVIEW-BEFORE-COMMIT FLOW

- 1.
- 2.
- 3.
- 4.

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1. Render model output in an editable structured form.
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REVIEW-BEFORE-COMMIT FLOW

1. Render model output in an editable structured form.
2. Highlight uncertain or policy-sensitive fields clearly.
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2. Highlight uncertain or policy-sensitive fields clearly.
3. Require explicit confirmation before any high-impact action.
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The user's confirmation is part of the audit trail, not just the UX.

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- Mask sensitive fields by default where required by policy.
- Control field visibility by role and permission level.
- Prevent accidental copy or export of protected data.
- Apply redaction behaviour consistently across all components, not per screen.

STREAMING UX WITH SSE

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- Handle stream interruption with a clear fallback state.
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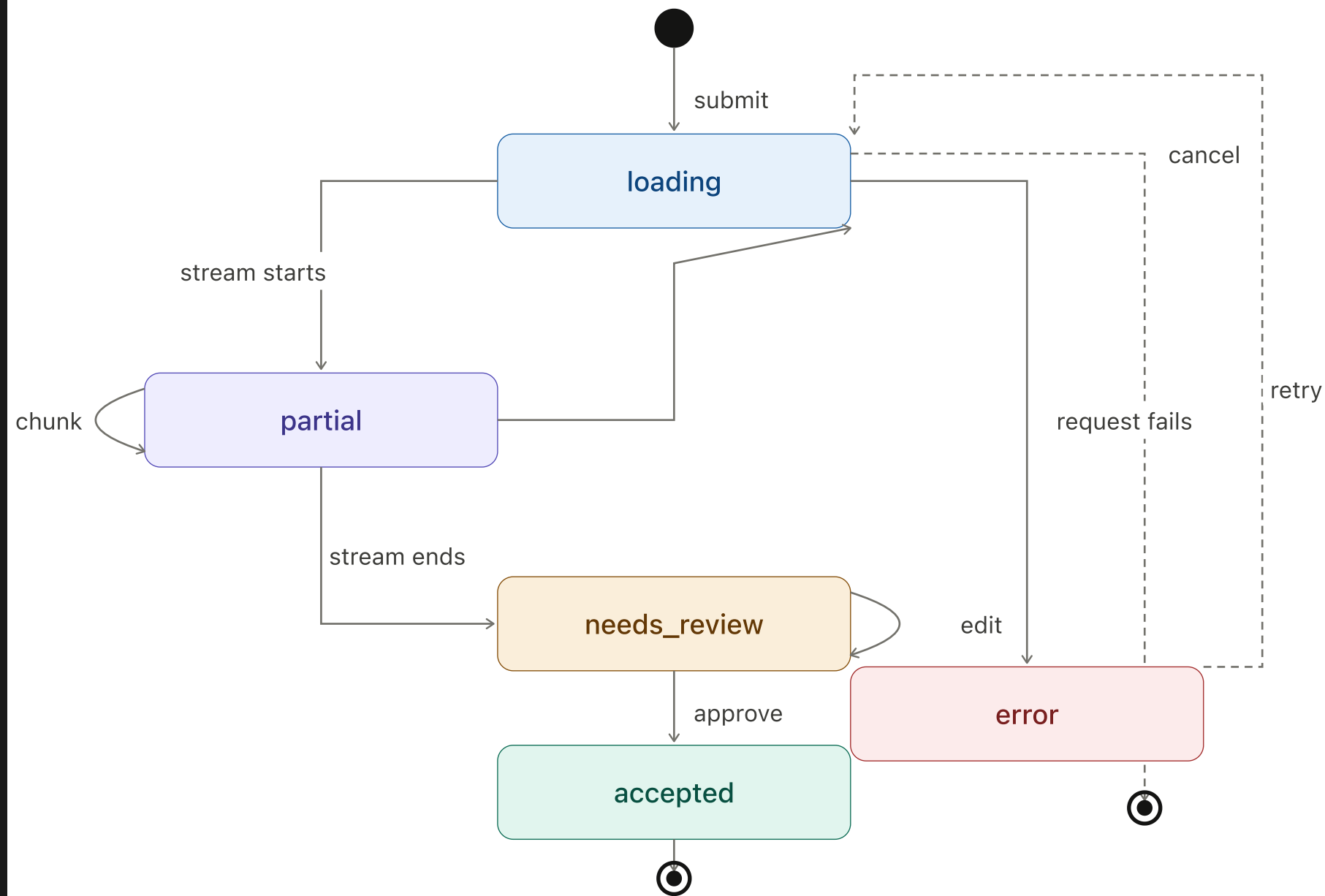
STREAMING UX WITH SSE

- Use deterministic state stages: queued, processing, partial, complete.
- Render partial updates without layout shift.
- Handle stream interruption with a clear fallback state.
- Never present partial output as final output.

STATE MODEL FOR RELIABLE UX

UI state	Meaning	User action
loading	Request in progress	Wait or cancel
partial	Stream incomplete	Monitor progress
needs_review	Uncertain or policy-sensitive output	Review, edit, approve
accepted	Validated, safe output	Continue workflow
error	Request failed	Retry or escalate





Every state must have a defined user action. Undefined states become support tickets.

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- One-click actions on unreviewed model output
- A single confidence number with no context or band definition
- Inconsistent fallback behaviour between screens
- UI state contracts that diverge from backend response statuses

UX TEST SCENARIOS

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UX TEST SCENARIOS

1. High confidence output with valid supporting evidence.
2. Low confidence on one critical field requiring review.
3. Policy-blocked field requiring redaction before display.
4. Interrupted SSE stream at partial completion.
5. Fallback response from backend returning `needs_review`.

MODULE 5 LAB BUILD TARGET

Build a TanStack component that:

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- Handles streaming states and fallback states explicitly
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Build a TanStack component that:

- Displays structured extraction output with field-level status
- Shows confidence bands and supporting evidence
- Supports an edit-before-save review flow
- Handles streaming states and fallback states explicitly
- Applies safe display and redaction where policy requires it

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Definition of done: every UI state from the state model has a defined, tested rendering path.

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SUMMARY

1. **UX is part of governance:** it shapes what decisions users make and how.
2. **Structured UI beats generic chat** for production workflows that require verification.
3. **Uncertainty must be visible** and designed to be acted on, not hidden.
4. **State discipline** is essential for streaming, fallback, and audit reliability.

BRIDGE TO MODULE 6

What we have now:

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What is next:

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- Your first task in Module 6: define what "correct" looks like for your extraction feature using a golden dataset.

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- Good UX needs measurable quality behind it to be trustworthy over time.
- Your first task in Module 6: define what "correct" looks like for your extraction feature using a golden dataset.

Module 6 covers evaluation and quality assurance: repeatable test sets, pass/fail criteria, and prompt

QUESTIONS?

Module 5 — Governed AI Feature Delivery